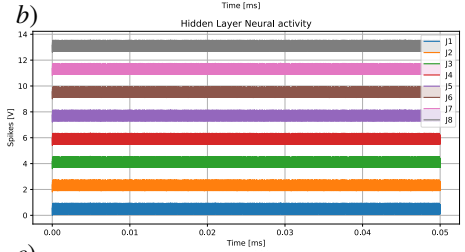
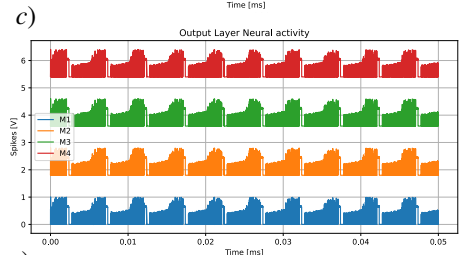
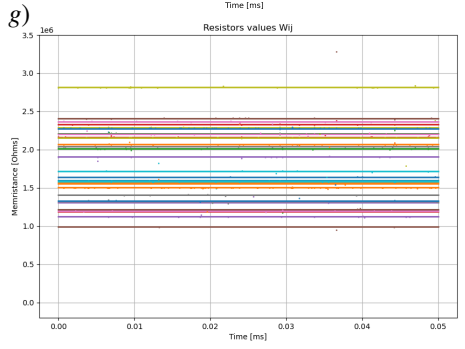
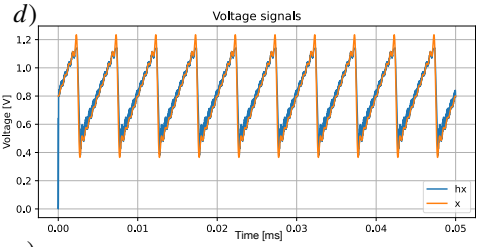
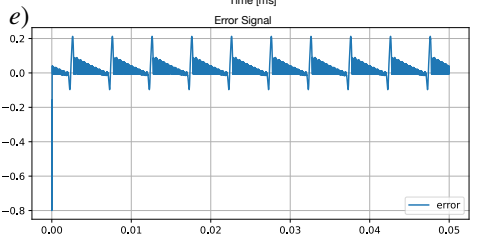

a)
Input Layer Neural activity
N1
N2
N3
N4
Spikes [V]
Time [ms]

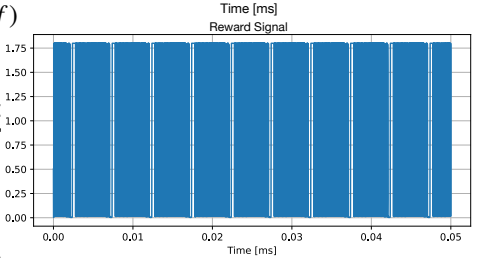

b)
Hidden Layer Neural activity
J1
J2
J3
J4
J5
J6
J7
J8
Spikes [V]
Time [ms]

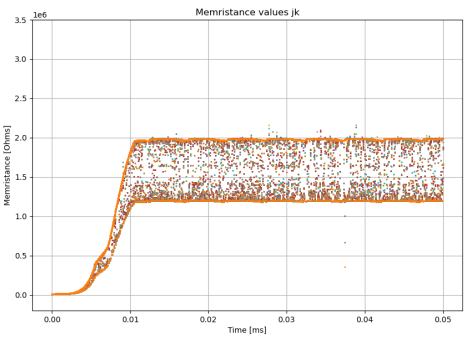

c)
Output Layer Neural activity
M1
M2
M3
M4
Spikes [V]
Time [ms]


g)
Resistors values Wij
1e6
Memristance [Ohms]
Time [ms]


d)
Voltage signals
Voltage [V]
hx
x
Time [ms]


e)
Error Signal
Voltage [V]
error
Time [ms]


f)
Reward Signal
Voltage [V]
Time [ms]


h)
Memristance values jk
1e6
Memristance [Ohms]
Time [ms]